Rohan Pedada

B.Tech Electrical Engineering, National Institute of Technology Mizoram | Roll No.: BT23EE006
+91 9032900219 bt23ee006@nitmz.ac.in pedadarohan@gmail.com
github.com/pedadarohan-dot linkedin.com/in/rohan-pedada

ABOUT

Software-focused AI/ML Engineer and Electrical Engineering undergraduate with a 9.34 CGPA and hands-on experience building **Generative AI, RAG, machine learning, computer vision, automation, and API-driven applications**. Experienced with **Python, JavaScript, data structures and algorithms, REST APIs, NLP, retrieval workflows, data preprocessing, model evaluation, debugging, testing, and optimization**. Built AI-powered workflows including **RAG chatbots, AI agents, recommendation systems, and API-integrated automation systems** using n8n and Python. Strong interest in **Agentic AI, LLM applications, multi-step workflows, knowledge retrieval, AI evaluation, and production-oriented software engineering**.

EDUCATION

National Institute of Technology Mizoram **2027**
B.Tech in Electrical Engineering; CGPA: 9.34 till 6th semester Mizoram, India

Narayana Junior College **2023**
Class XII, Andhra Pradesh Board of Intermediate Education; 97.7% Andhra Pradesh, India

Geetanjali School **2021**
Class X, Andhra Pradesh Board of Secondary Education; 100% Andhra Pradesh, India

TECHNICAL SKILLS

Programming	Python, C, JavaScript, HTML, CSS
Core CS	Data Structures & Algorithms, Algorithm Design, Object-Oriented Programming, Problem Solving, Debugging, Performance Optimization, Software Engineering, System Design
AI/ML	Generative AI, NLP, Machine Learning, Computer Vision, Recommendation Systems, Model Evaluation, Feature Engineering, Predictive Modeling
GenAI	RAG, Retrieval-Augmented Generation, AI Agents, Agent Workflows, Prompt Engineering, Retrieval Quality Evaluation, Workflow Automation, Knowledge Retrieval
APIs & Backend	REST APIs, API Integration, Service Integration, Backend Workflows
Libraries	NumPy, Pandas, Scikit-learn, PyTorch, OpenCV, MediaPipe
Tools	Git, GitHub, Linux/Bash, Docker, n8n, MATLAB
Systems	Distributed Systems, Event-Driven Systems, Automation, Testing, Regression Testing, System Monitoring

INTERNSHIP EXPERIENCE

Satish Dhawan Space Centre (SDSC-ISRO, SHAR) **17 June 2026 – 16 July 2026**
Student Project Intern *Sriharikota, Andhra Pradesh*

- Worked on **servomotor commissioning using PLC and servodrive**, applying control, debugging, testing, and industrial automation concepts to a real-world engineering system.
- Developed an **automatic fluid mixer control system using PLC and SCADA** for process automation and monitoring.
- Integrated **PLC programming, SCADA monitoring, and servo control** to implement and test an automated process-control workflow.
- Applied systematic **debugging, testing, monitoring, and problem-solving** techniques during automation system implementation.

IIT Guwahati **18 May 2025 – 18 June 2025**
Research Intern, Applications of AC and DC in Welding *Guwahati, Assam*

- Researched applications of AC and DC in welding under the guidance of Prof. Uday Shanker Dixit.
- Compared AC and DC behavior in welding applications and documented the technical differences between both systems.

Rashtriya Ispat Nigam Limited, Visakhapatnam Steel Plant **02 December 2024 – 28 December 2024**
Study Project Intern, Power System Configuration of Thermal Power Plant *Visakhapatnam, Andhra Pradesh*

- Studied power systems in the Thermal Power Plant department under the guidance of Krishna Kumar, HOD, Thermal Power Plant Department.
- Analysed the complete generation-to-distribution workflow and the major power system equipment used from generation to transmission.

PROJECTS

- **CineWatch (High-Scale Recommendation Engine):** Built an end-to-end **hybrid recommendation system** combining TF-IDF content-based filtering, SVD-based collaborative filtering, and transformer-based emotion classification. Processed **32M+ MovieLens ratings** for collaborative recommendation and integrated NLP-based emotion classification to generate mood-aware movie recommendations. Applied data preprocessing, feature engineering, machine learning, NLP, model evaluation, and iterative optimization to improve recommendation quality. Developed the system as a scalable AI application with an end-to-end workflow from data processing and model development to inference and recommendation generation.
- **Self-Calibrating Monocular Metric Metrology Engine:** Developed a real-time computer vision system for estimating real-world distances and dimensions from a single webcam. Implemented **MediaPipe landmarks, PnP head-pose estimation, Kalman filtering, camera calibration, and rotation-invariant contour-based measurement**. Combined computer vision algorithms, geometric estimation, filtering, and calibration into an end-to-end real-time measurement pipeline. Debugged and optimized the processing workflow to improve measurement stability and real-time performance.
- **SVM-Based Predictive Failure Maintenance:** Developed an end-to-end machine-failure prediction pipeline using **Support Vector Machines (SVM)** on the AI4I predictive-maintenance dataset containing 10K observations. Performed feature selection, stratified data splitting, standardization, model training, and evaluation using Python and Scikit-learn. Compared Linear, Polynomial, and RBF kernels and applied **5-fold cross-validation and grid-search hyperparameter tuning** to optimize model performance. Achieved **95.2% accuracy and 54.29% F1-score** with the optimized RBF model and reproduced the workflow in MATLAB. Used systematic experimentation and evaluation to compare model behavior and select the final approach.
- **AI Agents & Automation Workflows :** Designed and developed practical **AI-powered automation workflows**, including a Daily News Agent, Tools Finder Agent, AI RAG Chatbot, and other AI applications. Integrated **APIs, n8n workflows, automated tool execution, and multi-step workflow logic** to automate repetitive tasks and connect AI systems with external services. Tested and iteratively refined **prompts, retrieval workflows, response relevance, and system behavior** to improve reliability and output quality. Applied debugging and experimentation to identify workflow issues and improve AI application behavior.

- **RAG Chatbot:** Developed and tested a **Retrieval-Augmented Generation (RAG) chatbot** using n8n workflows and APIs for knowledge retrieval and AI-generated responses. Designed retrieval and workflow logic to provide relevant context to the generation process and improve response grounding. Evaluated **retrieval quality, response relevance, accuracy, and system behavior** through iterative testing. Refined prompts, retrieval workflows, and application logic based on evaluation results to improve overall response quality and reliability.
- **Smart Renewable Energy Optimizer (SREO):** Built an **AI-driven renewable energy optimization system** for efficient load distribution and energy utilization using **Python and data-driven optimization techniques**. Applied analytical and optimization approaches to develop an energy-management workflow focused on efficient utilization of renewable energy resources.
- **Telco Customer Churn Prediction Model:** Developed a **machine learning model for telecom customer churn prediction** using **Python, Pandas, NumPy, and Scikit-learn**. Performed **data preprocessing, feature engineering, model training, and model evaluation**, implementing and evaluating **Logistic Regression and Random Forest** approaches for churn prediction.
- **Handwriting Recognizer:** Implemented a handwriting recognition system using **Python, Scikit-learn, and machine learning techniques** to process handwritten input and convert it into digital text.
- **Campus Way:** Developed an AI-powered campus navigation platform with intelligent routing and assistant functionality using **JavaScript and Leaflet.js**. Implemented navigation functionality and integrated AI-assisted development workflows to build and iterate on a practical software application. Applied software engineering and problem-solving practices to develop and refine application features.

ADDITIONAL PROJECTS

- **Electrical Projects:** Automated Firing Angle Control, Automated Fluid Mixer Control and Commissioning of Servodrive
- **Developer Tools:** File Converter, Code Converter, SEO Meta Tag Analyzer.
- **Web Applications:** Habit Tracker built with HTML, CSS, JavaScript, and rapid prototyping tools.

CREDENTIALS AND ADDITIONAL INFORMATION

Key Courses	Power Systems, Electrical Drives, Electrical Machines, Control Systems
Certifications	Generative AI Mastermind – Outskill; AI for Engineers – Outskill; AI Tools and ChatGPT Expert – Be10X; Python 101 for Data Science – IBM
Leadership	Event Handler, Robotics Club, NIT Mizoram; Technical Writer, Electrical Engineering Society, NIT Mizoram
Achievements	Secured 1st position in the UI/UX competition and 3rd position in Samyug Hackathon at Technox Hackathon 2026
Languages	English, Hindi, Telugu
Interests	Technology, Chess